Höhere Technische Bundeslehranstalt Wels

4600 Wels, Fischergasse 30

ELECTROTECHNICAL LABORATORY

Title: **12. Resonance Circuit**

Date: 13.04.2017 / 20.04.2017

Author: Michael Gnadlinger

Group / Classname: A / 3AHME

Training supervisor:

Prof. Schwarzelmüller

Team members:

Grüblinger Lukas

Gnadlinger Michael

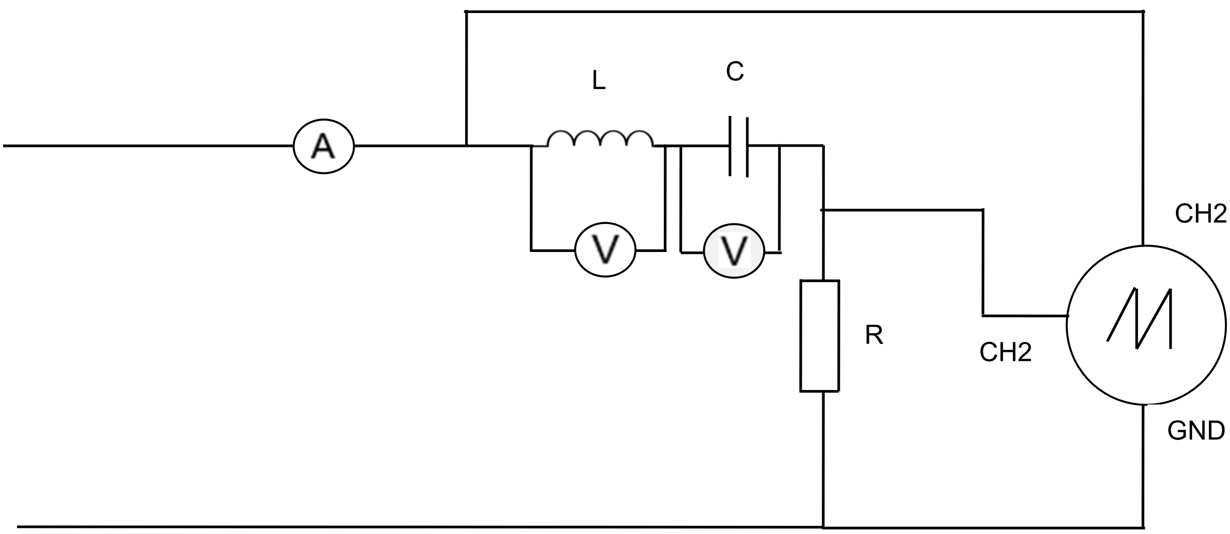
Used instruments:

Gwinstek GDS-1022 824-30-229; W1.17-224 - oscilloscope
Fluke 177 True RMS W1.17-70
Fluke 177 True RMS W1.17-72
Fluke 177 True RMS W1.17-73
2x 1000 Ω plug-in resistor
coil with 1200 windings
capacitor with 4.7µF
Agilent U1732A W1.17-088 – CLR measuring instrument
Gwinstek AFG-2105 W1.17-326 – function generator

Remarks:

12.1 Series Resonance Circuit:

Preparation:



oscilloscope: Gwinstek GDS-1022 824-30-229; W1.17-224

V_L-Meter: Fluke 177 True RMS W1.17-73
V_C-Meter: Fluke 177 True RMS W1.17-70
A-Meter: Fluke 177 True RMS W1.17-72

R_m (R_{Shunt}): 10 Ω-slider resistor
X_L: coil with 600 windings

R = 505.8 Ω (2x 1000 Ω in parallel) |
L = 5.414 mH (400 windings) - measured with Agilent U1732A W1.17-088
C = 4.773 µF |

$f_r = 1/[2*\pi*\sqrt{(L*C)}] = 990.07$ Hz

Ch1 = U_{ges}
Ch2 = I_{ges}

The task was to find a serial combination of a coil, a capacitor and a resistor to get a resonance circuit with a resonance frequenzy of around 1000Hz. Then supply it with a V_{pp} = 3V sin-wave from a function generator and measure U_{ges}, U_c, U_L, I_{ges}, Z and phse shift φ over Δt. Therefore we needed to rise the frequenzy logarithmical from 0 to 10 kHz (max. frequenzy of the multimeters). At the end we had to further investigate behavior at the resonance frequenzy.

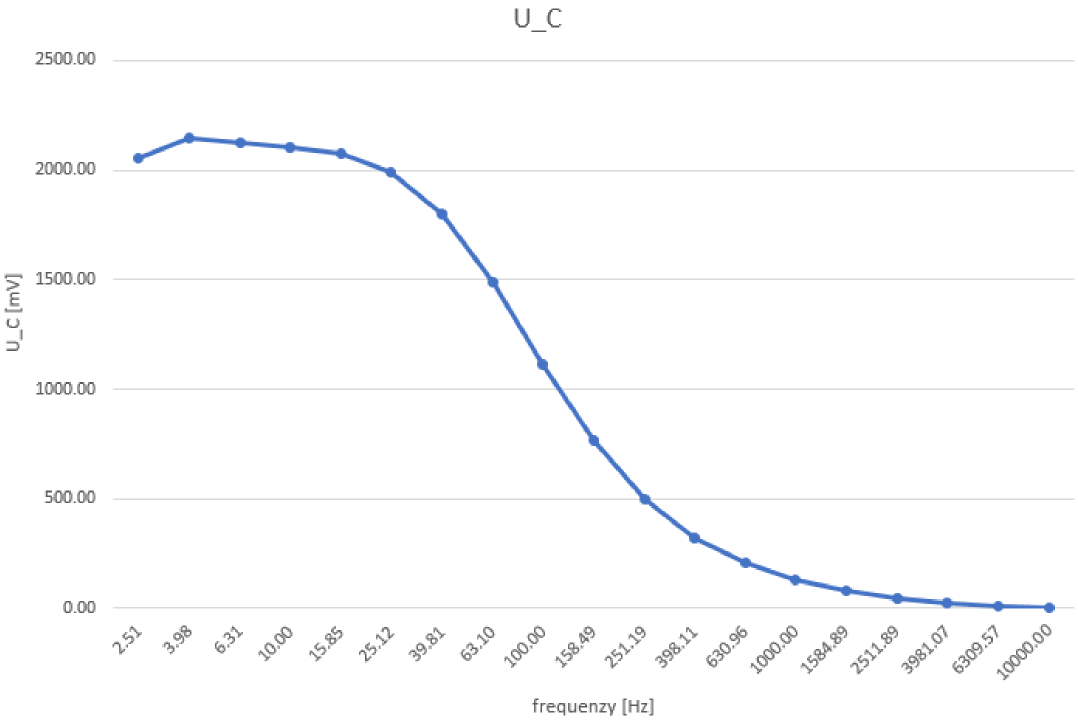
Measurement:

gegeben		gemessen				berechnet		Nebenrechnung	
[Hz]	[V]	[mV]	[mV]	[mA]	[µs]	[Ω]	[°]	[Ω]	[Ω]
f	**U_ges**	**U_C**	**U_L**	**I_ges**	**Δt**	**|Z|**	**ϕ**	**|Z_C|**	**|Z_L|**
1.58	3.00	17000.00	0.50	0.13	-156000.00	21110.96	-88.73	21104.95	0.05
2.51	3.00	2055.00	0.80	0.18	-96000.00	13294.73	-86.75	13285.19	0.09
3.98	3.00	2147.00	1.30	0.27	-60000.00	8393.47	-85.97	8378.35	0.14
6.31	3.00	2128.00	2.00	0.42	-37000.00	5308.54	-84.05	5284.60	0.21
10.00	3.00	2104.00	3.20	0.66	-23000.00	3372.39	-82.80	3334.58	0.34
15.85	3.00	2074.00	4.90	1.02	-13000.00	2163.26	-74.18	2103.84	0.54
25.12	3.00	1988.00	7.40	1.53	-7600.00	1419.76	-68.73	1327.46	0.85
39.81	3.00	1799.00	10.90	2.17	-3800.00	977.33	-54.46	837.62	1.35
63.10	3.00	1488.00	15.00	2.84	-2000.00	729.96	-45.43	528.46	2.15
100.00	3.00	1111.00	19.70	3.36	-880.00	603.96	-31.68	333.46	3.40
158.49	3.00	765.00	26.20	3.66	-340.00	545.77	-19.40	210.40	5.39
251.19	3.00	500.00	36.90	3.80	-132.00	520.83	-11.94	132.75	8.54
398.11	3.00	321.90	54.80	3.88	-29.00	510.65	-4.16	83.76	13.54
630.96	3.00	204.40	84.30	3.90	-14.00	506.77	-3.18	52.85	21.46
1000.00	3.00	128.30	130.70	3.90	0.00	505.80	0.00	33.35	34.02
1584.89	3.00	79.40	202.20	3.87	7.20	506.87	4.11	21.04	53.91
2511.89	3.00	47.60	305.80	3.80	9.20	510.92	8.32	13.28	85.44
3981.07	3.00	26.50	438.40	3.63	9.90	521.51	14.19	8.38	135.42
6309.57	3.00	12.40	568.40	3.27	9.90	547.41	22.49	5.28	214.63
10000.00	3.00	2.80	636.50	2.56	10.00	607.69	36.00	3.33	340.16

Caclulation of the measurements:

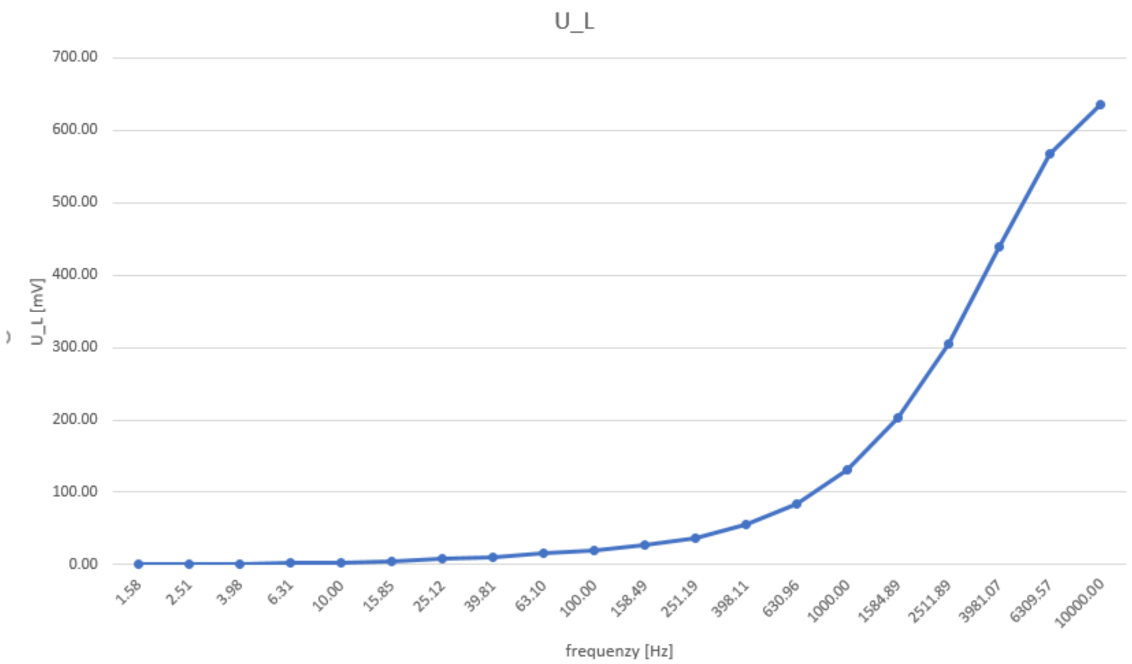
$$\varphi = {}^{\Delta t}/_{T} * 360°$$

$$|Z\_ges| = \left|\sqrt{(Z\_C - Z\_L)^2 + R^2}\right|$$



Conclusion:

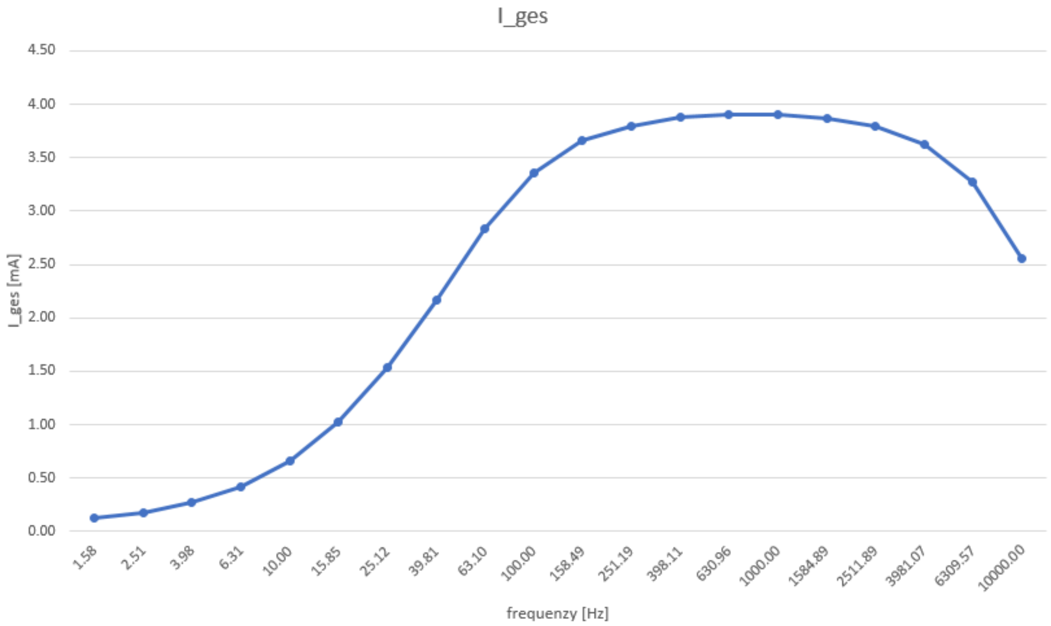
At low frequenzy the capazitor has a rather high resistance and therefore a higher voltage. This effect decreases when the frequenzy is increased.



Conclusion:

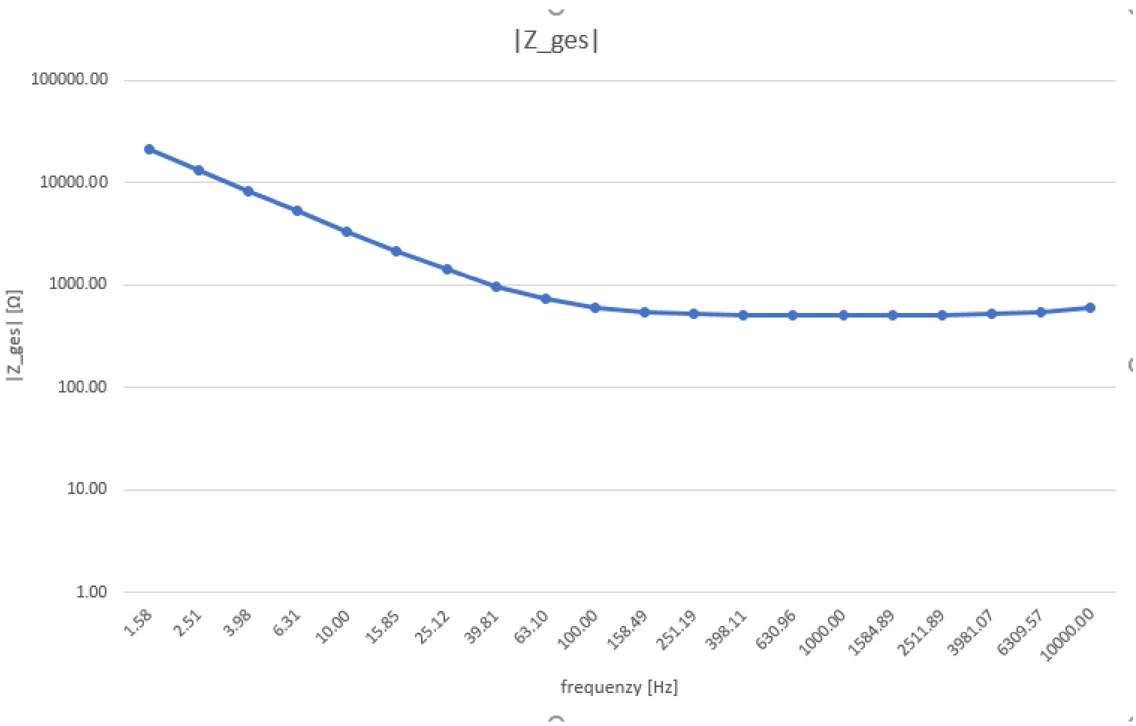
Other than the capacitor, the coil has a low resistance at a lower frequenzy and therefore a lower voltage, but a higher resistance when the frequenzy is increased.

You can also see, that at the resonace frequenzy, both components have approxamatelly the same resistance and therefore the same voltage drop of around 130 mV (as seen in the table).



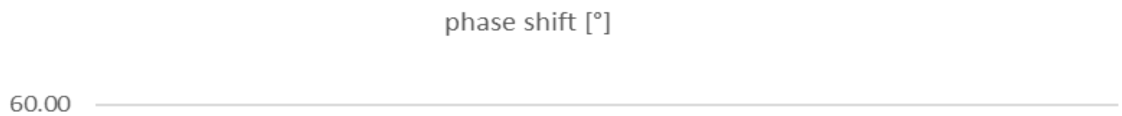
Conclusion:

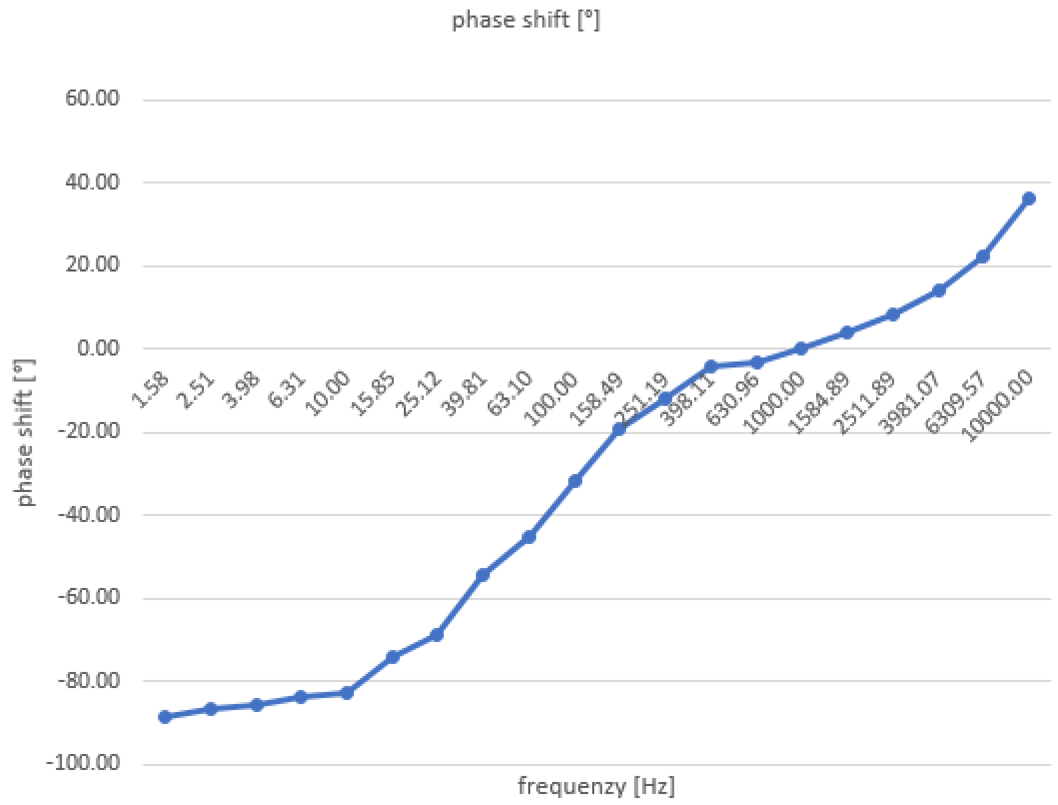
Because absolute value of the resistance of both compents is the same at the resonace frequenzy, but in the opposite direction of the imaginary-axis the left over resitance of our designed circuit is just the resistor R. This leads to a higher current than normal.



Conclusion:

Because the impedance of the coil doesn't increase at a higher frequenzy as much as the capacitor does at the lower frequenzy range and we used a rather big dampening resistor, the overall impedance at the resonance frequenzy isn't as segnificantly different than the impedance at the higher frequenzy. But it can still be seen that at 1000 Hz the only resistance of our circuite is just the 500 Ω dampening resistor.



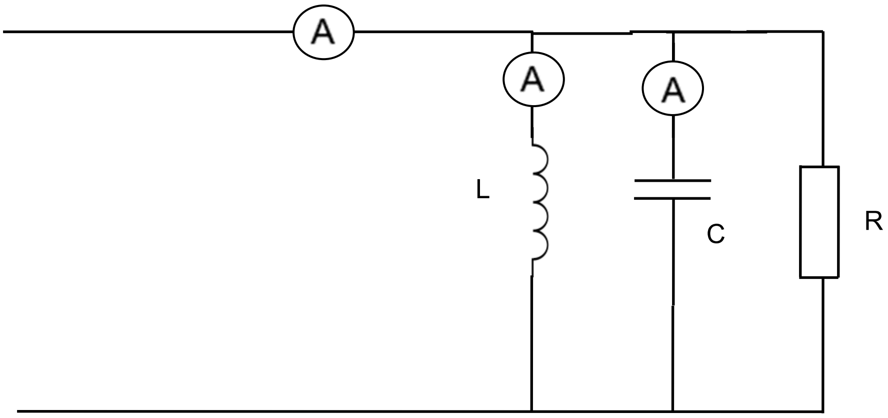


<u>Conclusion:</u>

I guess this diagram should look like a function 3rd grade with a saddle point at the resonance frequenzy which still can be seen. The cause of this bad graph may be a poor measuring job of us because to calcualate the phase shift we needed to measure the time difference with the cursor function of the oscillioscope which can get a bit inaccurate.

12.2 Series Resonance Circuit:

Preparation:



A_L-Meter: Fluke 177 True RMS W1.17-73
A_C-Meter: Fluke 177 True RMS W1.17-72
A-Meter: Fluke 177 True RMS W1.17-70

We used the same components as for task 12.1 and therfore got the same resonance frequeny of about 1000 Hz.
Also the supplied voltage was still 3V out of the same function generator and increasing the frequenzy logarithmical.

For this task only the currents at the coil, the capacitor and the overall circuit has to be measured simply using multimeters. Then the impedanze in dependence of the frequenzy has to be calculated.

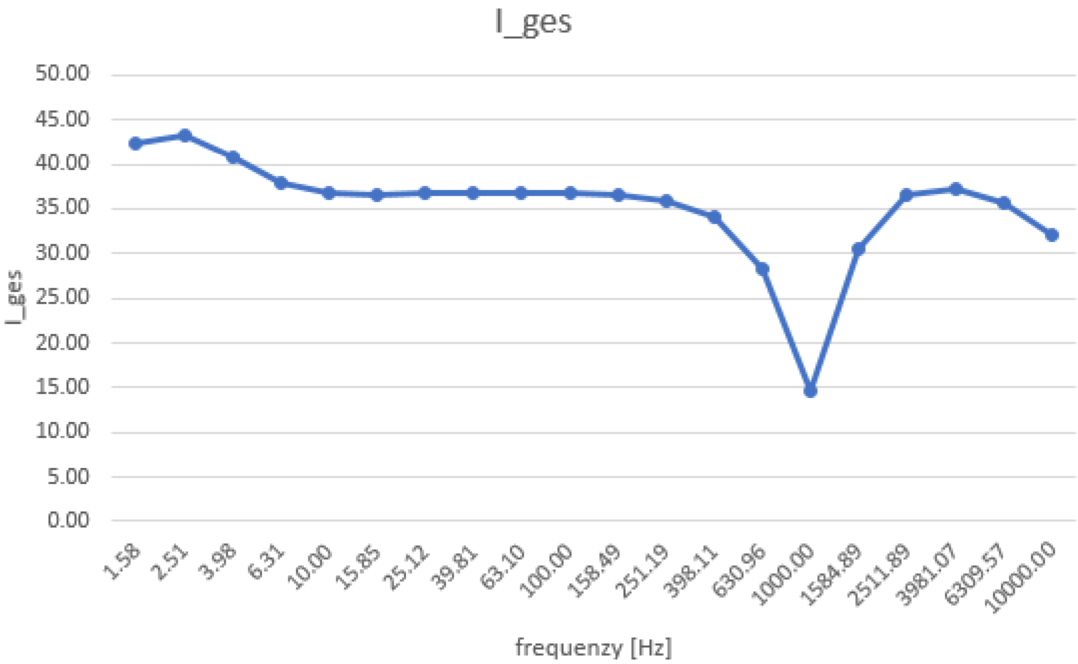
Measurements:

gegeben		gemessen			berechnet			Nebenrechnung	
[Hz]	[V]	[mA]	[mA]	[mA]	[Ω]	[°]		[Ω]	[Ω]
f	**U_ges**	I_ges	I_L	I_C	|Z|	ϕ		**|Y_C|**	**|Y_L|**
1.58	3.00	42.39	42.08	0.08	0.05	4.21354202		0.00	18.61
2.51	3.00	43.24	42.50	0.08	0.09	2.65306069		0.00	11.71
3.98	3.00	40.78	40.07	0.08	0.14	1.67332516		0.00	7.39
6.31	3.00	37.93	37.31	0.09	0.21	1.05546059		0.00	4.66
10.00	3.00	36.73	36.26	0.11	0.34	0.66596608		0.00	2.94
15.85	3.00	36.67	36.22	0.14	0.54	0.4201059		0.00	1.85
25.12	3.00	36.81	36.38	0.21	0.86	0.26497277		0.00	1.17
39.81	3.00	36.88	36.48	0.32	1.36	0.16703448		0.00	0.74
63.10	3.00	36.88	36.57	0.51	2.16	0.10512458		0.00	0.47
100.00	3.00	36.81	36.70	0.84	3.44	0.06592471		0.00	0.29
158.49	3.00	36.59	37.01	1.53	5.53	0.04094737		0.00	0.19
251.19	3.00	35.97	37.76	3.08	9.13	0.02480878		0.01	0.12
398.11	3.00	34.18	39.58	7.07	16.15	0.01402521		0.01	0.07
630.96	3.00	28.27	43.50	18.24	36.05	0.00626907		0.02	0.05
1000.00	3.00	14.62	41.29	42.12	484.61	0.00013389		0.03	0.03
1584.89	3.00	30.42	18.94	47.66	34.43	0.00656578		0.05	0.02
2511.89	3.00	36.45	6.87	42.66	15.71	0.01441499		0.08	0.01
3981.07	3.00	37.23	2.59	39.41	8.93	0.02537559		0.12	0.01
6309.57	3.00	35.77	0.95	36.44	5.42	0.04181354		0.19	0.00
10000.00	3.00	32.08	0.20	32.20	3.37	0.06727701		0.30	0.00

Calculation of the measurements:

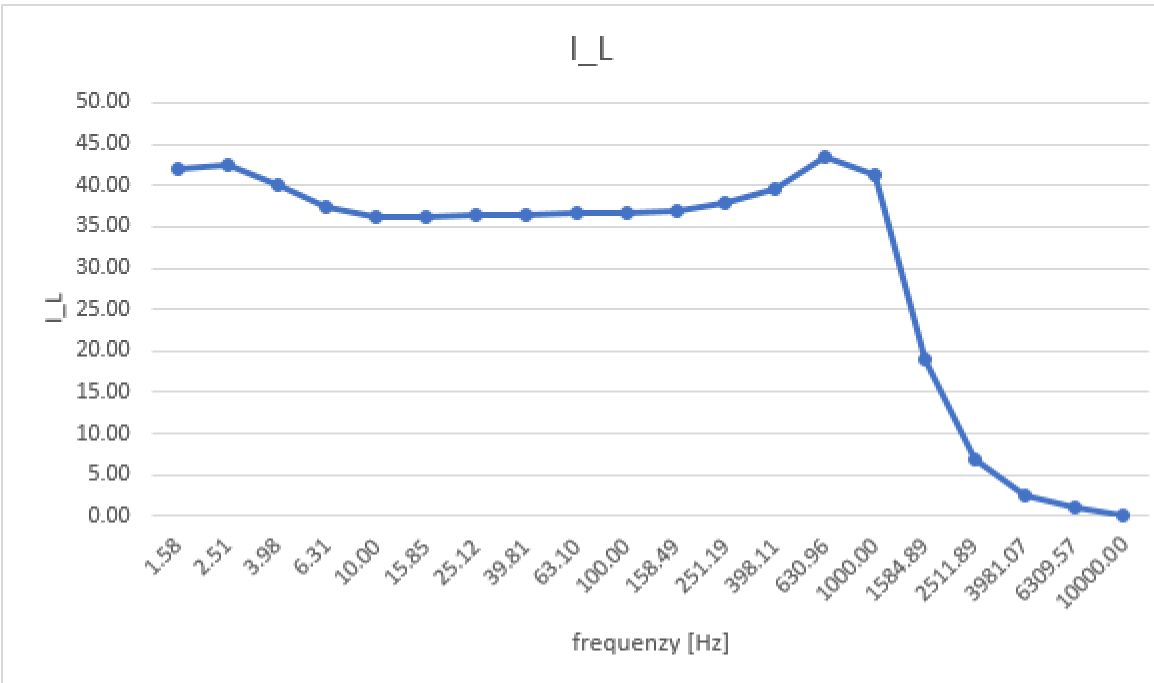
$$\varphi = \arctan \left({}^{(Y\_C - Y\_L)}\!/_{R} \right)$$

$$|Z\_ges| = \left|\sqrt{(Z\_C - Z\_L)^2 + R^2}\right|$$



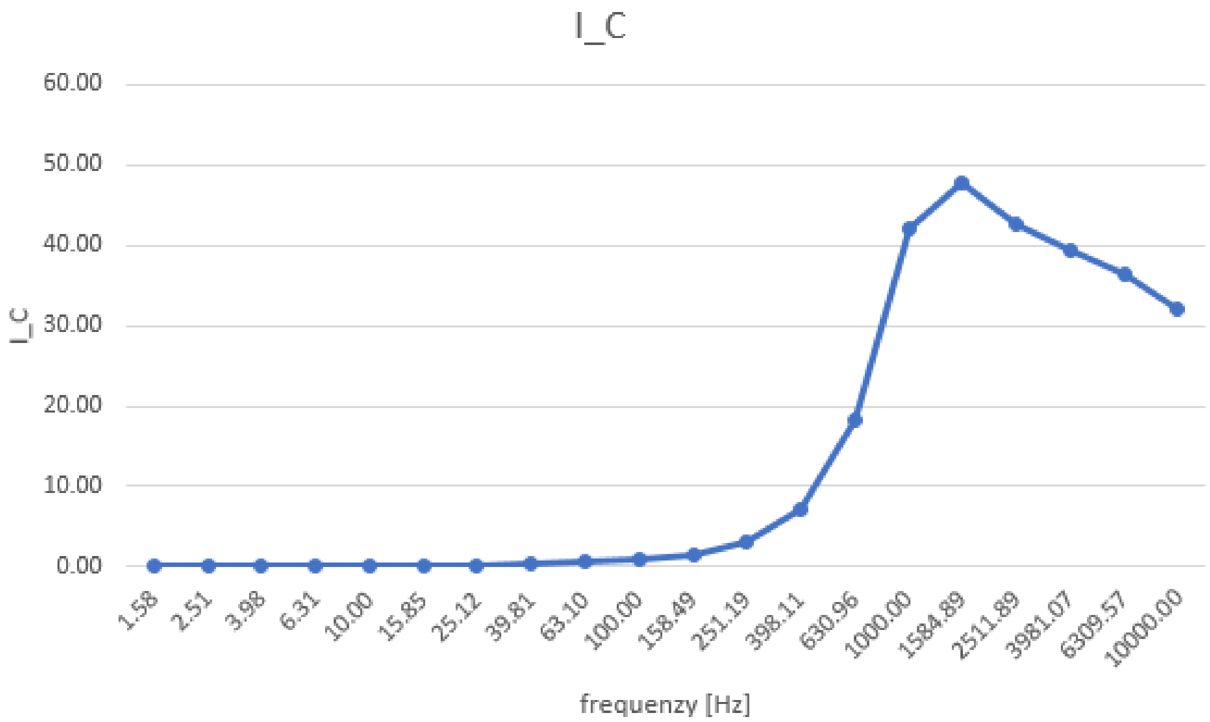
Conclusion:

As typical for a parallel resonance circiute the overall resistance increases heavily at the resonance frequenzy, other than the serial circuite, and therefore lowers the overall current it draws.



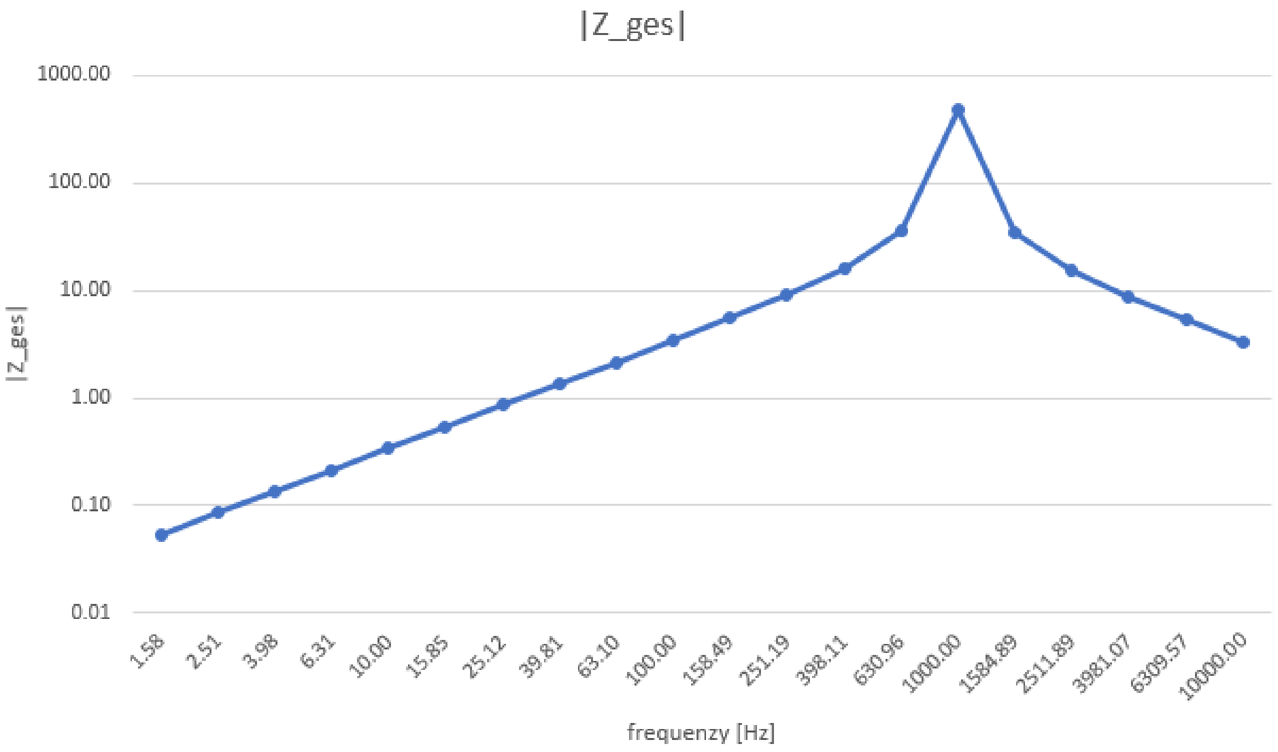
Conclusion:

It can clearly be seen that the admitance of a coil decreases with an increasing frequenzy, which ultimatlelly means a higher impendance. But it is interestin gto see that this effect just segnificantelly affects the current draw after the resonance frequenzy. This may be better explained with the next diagram.



<u>Conclusion:</u>

On the one hand the capacitor has a very high impedance at the lower frequenzy range which means that it heavily affect the current flow and almost allows no current to pass through the capacitor. On the other hand we used a fairly high resistor R of around 500 Ω. Ultimatelly this means that as long as the capacitor doesn't draw seginificantelly more currnet the the coil increases it's impedance with the frequenzy almost all of the current will flow throught it, with only little affect of the frequenzy.
In this diagram it can be observed that the capacitors impedance decreases with the frequqnzy.



Conclusion:

Due to the logarithmical scale of the impedance it seems to increase linear but only by a little bit. This keeps going until near the resonance frequenzy where the overall impedance peaks out to decrease again. This peak is because the apedances of the coil and the capacitor are of the same absolute value but in opposite directions which means the impedance is theoratically just covered by the resistor R. In our setup we didn't quite get this result: the impedance at the resoance frequenzy was 484.61 Ω.